Thijmen **Nijdam**

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Education

MSc Artificial Intelligence **Sep. 2023 - Feb. 2026**

University of Amsterdam

- Grade: 8.3
- Relevant courses: Natural Language Processing, Computer Vision, Information Retrieval I & II, Foundation Models, ML I & II, DL I & II
- Thesis: Built a world model to predict object dynamics for robotic manipulation, achieving zero-shot generalization to unseen objects

BSc Artificial Intelligence **Sep. 2020 - Jun. 2023**

University of Amsterdam

- Grade: 8.1, graduated with honours
- Minor in computer science, exchange at the University of Hong Kong

Experience

Robot Learning Research Engineer, *University of Amsterdam* **May 2026 - Present**

- Working on sim-to-real transfer for robot manipulation, enabling policies trained in simulation to generalise robustly to the real world

Research Intern, *University of Amsterdam* **Jan. 2024 - Feb. 2026**

- Benchmarked 40+ information retrieval models including LLM re-rankers and fine-tuned models to evaluate generalizability
- Benchmarked video generation models on physical reasoning
- Trained 3D diffusion models for radiotherapy treatment planning
- Implemented a novel conformal prediction method for time series decomposition

AI Developer, *LeerLevels* **Apr. 2022 - Mar. 2026**

- Work closely with founders to align deliverables with user needs
- Built RAG system for an assistant helping teachers assemble lessons
- Built AI assistant for editing educational content including front-end interface enabling teachers to review and apply suggestions
- Deployed an LLM-powered grading feature with monitoring via Langfuse, providing tailored feedback to hundreds of students

Publications

Benchmarking Physical Reasoning of Video Generative Models with Real Physical Experiments **ICML**

International Conference on Machine Learning 2026

- Built a benchmark for physical reasoning in video generation models with segmentation and object tracking pipelines
- Paper | Website

Reproducing NevIR: Negation in neural information retrieval **SIGIR**

48th International ACM SIGIR Conference on Research and Development in Information Retrieval 2025

- Benchmarked 40+ information retrieval models including LLM re-rankers and fine-tuned models to test generalizability
- Paper | Code | Poster

HIVE: A Hyperbolic Interactive Visualization Explorer for Representation Learning **ICCV Workshop**

2nd Beyond Euclidean Workshop: Hyperbolic and Hyperspherical Learning for Computer Vision 2025

- Built an interactive visualization tool for exploring hyperbolic embeddings
- Paper | Code | Demo

Conformal time series decomposition with component-wise exchangeability **COPA**

13th Symposium on Conformal and Probabilistic Prediction with Applications 2024

- Developed a decomposition-based conformal prediction approach for uncertainty quantification in time series forecasting
- Paper | Code | Slides

Reproducibility Study Of Learning Fair Graph Representations Via Automated Data Augmentations **TMLR**

Transactions on Machine Learning Research 2024

- Reproduced and extended a fair graph learning model, evaluating fairness–accuracy trade-offs on node classification and link prediction
- Presented at NeurIPS 2024 as part of the Machine Learning Reproducibility Challenge
- Paper | Code | Poster | Slides

Extracurricular Activity

AI Safety Reading Group Facilitator

- Facilitated 8 reading group sessions on AI alignment, preparing materials and guiding discussions 2024

Hackathons

- AI Safety Hackathon, focused on improving LLM interpretability, achieving 2nd place 2024
- European Robotics Forum Hackathon, implemented autonomous driving and object detection on a wheeled robot 2022